

Topological Segmentation Error and Boundary-Condition Tuning in Computed Coronary FFR: A Controlled In Silico Study

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Abstract—Fractional flow reserve, the pressure ratio that decides whether a coronary narrowing is treated, can be computed from computed tomography angiography. The computation depends on the segmented artery and on outlet boundary conditions derived from it, often tuned to measured perfusion, as in a digital twin. Whether a perfusion-matched model can still assign the wrong treatment at 0.80 is untested. We inserted 150 stenoses into 108 coronary trees, applied four segmentation error types (missed side branch, vessel break, longer lesion, narrowed taper), and recomputed fractional flow reserve with a reduced-order model under fixed, re-derived and perfusion-tuned boundary conditions, with one global or one per-territory tuning parameter, in two microvascular bed structures. With fixed boundary conditions, branching (topological) errors changed the decision in 32–33% of models, against 6–10% for vessel-size (caliber) errors of inter-observer magnitude. Re-derived or tuned boundary conditions reduced topological decision changes to 5–19%. In this threshold-stratified cohort, tuned models still passed a perfusion check stricter than measurement repeatability while wrong by more than 0.05: 19–20% (discrete bed) and 5–8% (leaky bed) of topological-error models and 4–18% of caliber-error models, mostly taper, against 3–4% for correct anatomy and none for caliber errors with re-derived boundary conditions. Per-territory tuning corrected the typical missed branch, and prescribed territory flows in three dimensions reduced a missed-branch shift of 0.074 to -0.0007 . The directions held at higher demand. A perfusion match after tuning therefore does not show that the computed value is correct; branching and caliber should be checked before tuning.

Index Terms—Boundary conditions, coronary artery disease, digital twins, fractional flow reserve, image segmentation, reduced-order modeling, uncertainty.

I. INTRODUCTION

FRACTIONAL flow reserve (FFR) is the pressure ratio that decides whether a coronary narrowing is treated. It is the blood pressure beyond the narrowing divided by the aortic pressure during maximal flow. A value at or below 0.80 marks a stenosis for revascularization [1]. FFR can

also be computed without a pressure wire. The lumen, the blood-filled channel of the artery, is segmented from coronary computed tomography (CT) angiography and converted into a flow model, and the computed FFR has been validated against invasive measurement in multicenter trials [2], [3]. The calculation exists in three-dimensional (3D), reduced-order and machine-learning forms [4], [5], [6].

Every such model needs boundary conditions, the pressures, flows or resistances imposed where the model ends. They stand in mainly for the microvascular bed, the arterioles and capillaries that set how much blood each region of heart muscle draws. Its resistance is not measured but derived from the segmented lumen through scaling laws that relate vessel size to the flow it carries [7], [4]. It is increasingly tuned so that the model reproduces measured perfusion, the blood flow to each region of heart muscle, or measured flow splits [8], [9]. In a coronary digital twin, this tuning personalizes the physiology. The segmentation therefore enters the computed FFR twice, through the arteries and through the boundary conditions derived from them.

Segmentation error has been studied mainly as caliber error. Caliber is the inner diameter of a vessel, so a caliber error is a lumen drawn too wide or too narrow, or a narrowing drawn too long, while the branching is unchanged. Sensitivity analyses perturb the radius at fixed bifurcations and report a continuous change in the computed pressure index [10], [11], [12], [13]. Studies that set geometry against boundary conditions rank the two by their influence on a continuous output [14], [15], [16]. Topological error, a change in the branching structure itself such as a missed side branch or a vessel that ends too early, has received less attention. In a pulmonary network model, connectivity contributed more to hemodynamic uncertainty than vessel radius and length [17], and examples from a recent coronary benchmark showed vessel breaks in the output of every evaluated method despite high overlap scores [18], and small distal segments remain the hardest to segment [19]. A missed branch removes part of the microvascular bed from the model and changes both the flow through the stenosis and the boundary conditions derived from the tree.

Whether such an error reaches the computed FFR depends on how the boundary conditions are handled. With outlet resistance held fixed, adding a side branch beyond a stenosis lowered FFR by 15.5% in an idealized model and by 13% in one of two models built from optical coherence tomography [20]. When a cohort-wide outlet resistance was recalibrated for a model with side-branch outflow, diagnostic accuracy changed

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little [21]. In a clinical cohort, a change of outlet model moved sensitivity from 58.1% to 68.6% at an unchanged area under the curve [5]. If the boundary conditions are tuned until the model matches measured perfusion, a topological error may vanish from the perfusion check yet remain in the pressure along the vessel. None of these studies compares fixed, re-derived and tuned boundary conditions on the same corrupted anatomy at 0.80, or tests whether a model can pass a perfusion check and still assign the wrong treatment.

We propose a controlled test for this problem, an ablation in which one segmentation error at a time is introduced into an otherwise correct model, whose FFR serves as the true value, so every change in FFR can be traced to the error and to the handling of the boundary conditions. The approach is new in three respects: it judges segmentation error by the treatment decision at 0.80 against the variability of repeat invasive FFR; it solves the same corrupted anatomy under fixed, re-derived and tuned boundary conditions, with one global or one per-territory tuning parameter, and separates topological from caliber errors of inter-observer magnitude; and it counts how often a model passes a perfusion check while its FFR is materially wrong, with a 3D computational fluid dynamics (CFD) case as a check. The results show what a perfusion check misses and which further checks a computed-FFR pipeline needs.

II. METHODS

In outline, we inserted a stenosis into each real coronary tree, introduced one segmentation error at a time (Fig. 1), recomputed FFR under three boundary-condition protocols and two bed structures, and compared each result with the clean model of the same tree.

A. Data and Cohort

ImageCAS-X provides voxel-wise lumen masks, labeled centerlines and surfaces for 800 CT angiography scans of the ImageCAS dataset [22], [18], [23]. We used its 160-scan test split, in which a second analyst re-annotated each scan. The two annotators agreed to a Dice similarity coefficient (DSC, the overlap of two masks) of 92.8% and a 95th-percentile Hausdorff distance (HD95, the surface distance exceeded by 5% of boundary points) of 2.46 mm [18]. Each scan gives a left and a right coronary tree. The radius at each centerline point is the Euclidean distance from the point to the nearest background voxel of the mask.

The dataset contains few lesions near the decision threshold, so we inserted one stenosis per instance into an otherwise unchanged tree. An instance is one lesion in one tree. A host vessel (left anterior descending, LAD; left circumflex, LCx; or right coronary artery, RCA) was eligible when image quality was at least adequate (2 on a 0–4 scale), the healthy reference radius r_{fit} at the lesion center (from a linear taper fit) was at least 1.0 mm, native narrowing was below 40% diameter stenosis (DS), the measurement point 20 mm beyond the lesion lay on a vessel of radius at least 0.75 mm, and the FFR before insertion was at least 0.90. The lesion is an axisymmetric

cosine narrowing (Supplementary Material),

$$r'(s) = \min(r, (1 - w)r + w r_{\text{fit}}(1 - \text{DS})), \quad (1)$$

$$w = \frac{1}{2} \left[1 + \cos \frac{2\pi(s-c)}{L} \right]$$

for $|s - c| < L/2$, where s is arc length, c the lesion center and L its length. The vessel was never widened. Lesions were centered 20 or 45 mm from the vessel origin, were 10 or 20 mm long and had 40, 50, 55, 60, 65, 70, 75 or 80% DS. Of the 6944 resulting instances (280 host vessels, 140 scans), we drew 25 in each of six 0.05-wide bands of baseline FFR from 0.65 to 0.95, at most two per tree, giving 150 instances (50 per host vessel) in 108 trees from 93 patients.

B. Reduced-Order FFR Model

All 150 instances were solved with a reduced-order model, which represents the coronary tree as a network of flow resistances instead of computing the full 3D flow field. The model places one node at every centerline point. Each element between consecutive nodes has the Poiseuille resistance $8\mu \Delta s / (\pi r^4)$ of its actual radius r and length Δs , with blood viscosity $\mu = 0.004$ Pas. Where the narrowing exceeds 30% DS, an expansion loss $\Delta p = K Q|Q|$ is added at the throat, with $K = \rho K_t (A_0/A_s - 1)^2 / (2A_0^2)$, $K_t = 1.52$ [24], density $\rho = 1060 \text{ kg m}^{-3}$, and A_0 and A_s the reference and throat areas. Aortic pressure is 90 mmHg and venous pressure 5 mmHg. The nonlinear equations are solved by under-relaxed fixed-point iteration, each step a sparse linear solve for the node pressures, until the relative flow change falls below 10^{-8} . The mass-conservation error is recorded for every solve. FFR is the computed pressure at the measurement point divided by aortic pressure. The model is steady and represents maximal hyperemia (maximal flow) without autoregulation.

The microvascular bed is modeled as conductances (the inverse of resistance) that drain flow from network nodes to venous pressure. Each conductance depends on a healthy reference radius r_{ref} , the taper-fit radius r_{fit} made non-increasing along each path, and is scaled by one bed constant C . Two bed structures were analyzed because they treat a deleted vessel differently. In the leaky bed, flow leaves along the vessel wall at every node of the tree truncated at $r_{\text{ref}} = 0.50$ mm, with conductance $(r_{\text{ref}}^3 - \sum r_{\text{ref,child}}^3) / C$, so that flow follows Murray's cube law [7]. This distributed outflow stands for the side branches the image does not resolve, and the conductance of a deleted branch reappears as wall outflow at its parent node. In the discrete bed, flow leaves only at the outlets of the tree truncated at $r_{\text{ref}} = 0.60$ mm, with conductance $r_{\text{ref}}^{2.66} / C$, where the exponent approximates 8/3, the inverse of the 3/8 power that relates coronary vessel diameter to the myocardial mass it perfuses [25], so that outlet flow is taken as proportional to perfused mass. The discrete bed is the outlet structure that a 3D model can represent. In both structures C was calibrated on the healthy-equivalent network, the tree with its reference radius and no lesion, so that inflow equals the hyperemic demand $Q = k r_{\text{in}}^3$, where r_{in} is the inlet radius and the cube law is Murray's [7]. With the model setting $k = 562 \text{ s}^{-1}$, a vessel with the normal proximal LAD diameter of 3.7 mm [26], carries 214 mL/min, within one standard deviation of

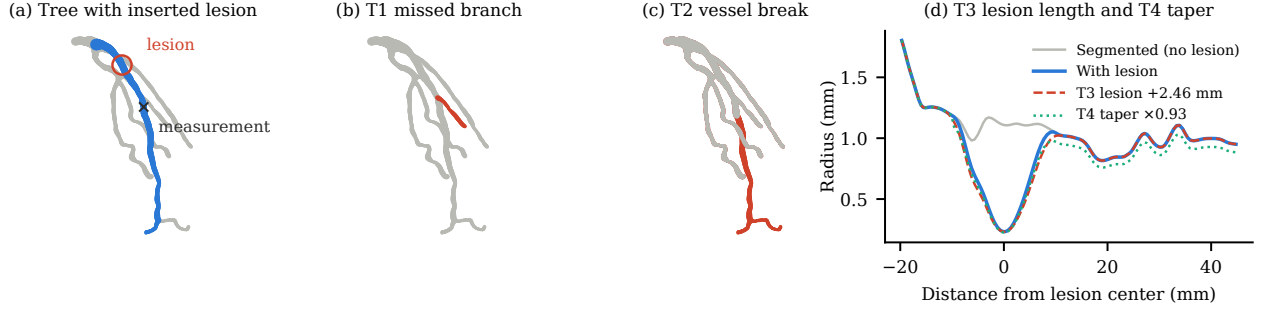


Fig. 1. The four segmentation error types on one cohort tree (left coronary tree of scan 14, front view, line width proportional to radius). (a) The tree with an inserted 80% DS lesion in the proximal LAD (blue) and the measurement point 20 mm beyond it. (b) T1 deletes the largest side branch beyond the lesion (red). (c) T2 ends the host vessel 25 mm beyond the lesion, and the red part is lost. (d) Radius along the LAD for the caliber errors, T3 lengthening the lesion by 2.46 mm and T4 shrinking the radius by 7% from the lesion onwards. The same instance was solved in 3D.

the hyperemic flow measured in that artery by continuous thermodilution (228 ± 71 to 293 ± 102 mL/min) [27]. The segmented arteries were narrower than these normal values (median inlet radius 1.60 mm), so the cohort's median demand was 2.3 mL/s (137 mL/min) per tree, and the full pipeline was repeated with k doubled and tripled (Section III-E). Reference radii and C are computed from the original tree, so an inserted lesion changes only the epicardial radius. The leaky bed was analyzed on all 150 instances. The discrete bed was analyzed on the 97 instances whose healthy-equivalent network gave a main-vessel FFR of at least 0.90 and had at least two outlets, and baseline FFR bands were derived separately for each bed. Halving the centerline spacing changed FFR by at most 0.005 in either bed.

C. Segmentation Error Types

Four error types were applied to each instance with its lesion in place. Two are topological errors, which change which vessels exist. Two are caliber errors, which change the lumen size or lesion extent while the branching is unchanged. The missed branch (T1) deletes the largest side branch that leaves the host vessel beyond the lesion, together with its subtree. The vessel break (T2) ends the host vessel 25 mm beyond the distal edge of the lesion, so the measurement point survives but the run-off beyond it is lost. The lesion length (T3) re-inserts the same stenosis 2.46 mm longer. The taper (T4) multiplies the radius by 0.930 from the proximal edge of the lesion onwards and in every descendant vessel. For T3 and T4 the set of modeled nodes is fixed to that of the clean tree, so a narrowed radius does not remove an outlet.

The caliber magnitudes are the inter-observer statistics of the test split, so each caliber error is as large as the disagreement between two annotators. HD95 (2.46 mm) sets T3. The DSC of 92.8% sets T4 as the radius ratio k of two coaxial cylinders, $2k^2/(k^2 + 1) = 0.928$. Because both annotators edited the same automatically generated centerlines, the dataset authors describe the inter-observer DSC as an upper bound on agreement [18]. T1 and T2 have no measured magnitude and are design choices. Nodes of the corrupted tree were matched to the clean tree by coordinate, and FFR was read at the same anatomical point.

D. Boundary-Condition Protocols

Each corrupted tree was solved under four protocols. Under Protocol A (fixed), surviving nodes keep the bed conductances and C of the clean model, and the bed of deleted vessels is lost. This represents the case in which the boundary conditions were obtained correctly despite the segmentation error. Under Protocol B (re-derived), the bed rule is applied to the corrupted tree and C is recalibrated to that tree's own hyperemic demand, as an automated pipeline does with whatever anatomy it receives. Under Protocol C (tuned), one global scaling of C (one bed scaling) is fitted so that the corrupted model reproduces the clean model's territory flows. A perfusion territory is the subtree below each child of the first bifurcation of the clean tree. Its target is the total bed outflow of that subtree in the clean model, including the share of any vessel the error deleted, because the heart muscle is perfused whether or not its artery was segmented. Each corrupted node belongs to the territory of its clean counterpart. The fit minimized the mean squared relative error of the territory flows on a 61-point grid over ± 1.5 decades of C , then refined it; a fit at the edge of the range was a failure. One parameter against two or more targets makes the fit over-determined, so this is the least flexible tuning a pipeline could use. Under Protocol D (per-territory tuned), the re-derived bed of each territory is scaled by its own factor, fitted (within $10^{\pm 3}$) so that every territory flow equals its clean target. With one parameter per target the fit is exactly determined, the most flexible tuning to territory flows. Protocols C and D are defined only when at least two territories retain a vessel. They were undefined for 33 of 77 (discrete) and 47 of 118 (leaky) missed-branch instances and for 36 of 96 and 49 of 149 vessel breaks, all but four in the RCA (a missed branch is possible only where a side branch leaves the host vessel beyond the lesion). Both match flow only, because matching both flow and pressure at the outlets of a steady tree would reproduce the clean FFR by construction.

For every protocol we computed the perfusion residual, the root mean square of the relative differences between the model's territory flows and the clean targets. It is the quantity a perfusion check compares with the measurement.

E. Three-Dimensional Case Study

To test whether the findings depend on the simplifications of the reduced-order model, one instance (a 20 mm lesion of 80% DS in the proximal LAD) was also solved with 3D CFD, which computes the full velocity and pressure fields from the Navier–Stokes equations by the finite-volume method. Three lumens were built: clean, with the lesion (baseline), and with the lesion and the missed branch (T1). Surfaces were extracted from the mask by marching cubes and Taubin smoothing; the lesion was applied by moving surface vertices radially with (1), and the branch was removed by deleting its voxels. Meshes of 3.5–3.9 million cells (cfMesh) had 25 μm cells within ± 4 mm of the throat and four boundary layers.

Steady laminar Newtonian flow, with the viscosity and density of the reduced-order model, was solved with the steady incompressible solver of OpenFOAM (simpleFoam, ESI v2406, SIMPLE pressure–velocity coupling), with a bounded second-order upwind scheme for convection and second-order central schemes elsewhere. Each solve ran 3 000 iterations; convergence required scaled residuals below 10^{-5} and a steady pressure at the measurement point. The walls were rigid with no slip, and the inlet carried aortic pressure. Each outlet either set its pressure to venous pressure plus the clean tree’s resistance times its flow (Protocol A), or received a prescribed clean territory flow, shared over the territory’s surviving outlets by bed weight (the per-outlet limit of Protocol D). FFR is the area-averaged pressure on a cross-section at the measurement point divided by aortic pressure. A grid-convergence study on an idealized 70% DS stenosis (throat cells of 50, 25 and 12.5 μm) gave a discretization uncertainty in FFR of 5.5×10^{-4} . A reduced-order counterpart of each geometry, rebuilt on the area-equivalent radius of the meshed lumen with the same outlet conditions, separates the effect of model fidelity from that of radius definition.

F. Outcomes and Statistical Analysis

For each instance, error type, protocol and bed, ΔFFR is the corrupted model’s FFR minus that of the instance’s own clean model in the same bed. A flip is a change of classification at 0.80 between the two, that is, a change in the treatment decision. A model passes the perfusion check when its perfusion residual is below 10%. Published repeatability of hyperemic myocardial blood flow (within-subject variation of 10–15% [28] and a regional repeatability coefficient of 30–37% [29]) implies a 95% bound of about 20–29% for one model output against one measurement, so the 10% threshold is strict. Thresholds of 13% and 16% were analyzed as sensitivities. An FFR is materially wrong when $|\Delta\text{FFR}| > 0.05$, an error large enough to matter near the 0.80 cut.

Proportions are reported with Wilson 95% confidence intervals (CIs). Protocol contrasts of flips (B against A, C against A, C against B) are paired within instance and tested with the exact McNemar test, with Holm adjustment across the three contrasts of each error type and bed; passes-and-wrong under tuning (C and D) is compared with Protocol B by the same test. Topological and caliber errors are compared by a paired sign test on the per-instance flip proportions of each class, and

paired changes in $|\Delta\text{FFR}|$ by the Wilcoxon signed-rank test. A Bayesian mixed-effects logistic model of flips with patient and lesion-slot random effects is reported in the Supplementary Material. Every analysis was run in both beds, effect sizes are reported per bed, and a finding is claimed only where its direction agrees in both.

Each flip rate is compared with a noise floor, the rate expected from measurement variability alone. For each instance, this is the probability that a repeat invasive FFR, drawn from a normal distribution centered on the clean value with the published standard deviation of the difference between repeat measurements, 0.018 [30], falls on the other side of 0.80, following the measurement-certainty approach of [31]. A second floor was simulated on the clean anatomy: 20 draws per instance perturbed cardiac output, aortic pressure, blood viscosity, the share of flow to each territory and the measured territory flows by their within-subject variability, and the model was tuned to the perturbed flows as in Protocol C.

The cohort, design, analysis plan and ablation code were fixed, and the cohort list hashed, before the ablation was run. The analysis script was written while the ablation ran and before its output was read. The threshold and demand sensitivities, noise floors and mixed model were specified in the plan; Protocol D was specified as a sensitivity but run after the primary results were read, and the demand replication was added afterwards. Because the cohort was stratified to span 0.80, its rates are conditional on that design and are not clinical prevalences. The analysis pipeline is shown in the Supplementary Material.

III. RESULTS

A. Decision Changes by Error Type and Protocol

Topological errors changed the decision more often than caliber errors under Protocols A–C in both beds (Table I). With fixed boundary conditions in the discrete bed, 45 of 137 topological-error models (33%, 95% CI 26–41%) crossed 0.80, against 20 of 194 caliber-error models (10%, 7–15%). In the leaky bed the proportions were 32% (26–38%) and 6% (4–9%) (paired sign test, $p \leq 0.002$ in both beds). Under Protocols B and C the difference kept its direction. It was significant in the discrete bed ($p = 0.013$ and 0.002) but not in the leaky bed ($p = 0.18$ and 0.057). Under Protocol D the two classes flipped equally often (8% and 11% discrete, 6% and 4% leaky). Flips were concentrated in the bands adjacent to 0.80 (Fig. 2). All 2 599 corrupted-model solves under Protocols A–C and 769 under Protocol D converged (maximum mass-conservation error 9×10^{-9}); in the discrete bed the vessel break removed every outlet in 36 instances under Protocol A.

The caliber errors rarely changed the decision. At the magnitude of inter-observer disagreement they changed FFR by a mean of 0.006–0.037 in absolute value, and their flip rates of 1–14% were of the order of the 5.0–6.5% expected from repeat invasive measurement. Only the taper exceeded this noise floor with its lower confidence bound, under fixed boundary conditions (13% discrete, 9% leaky) and under Protocol D in the discrete bed (14%). Against the grey zone of 0.75–0.85 [31], no caliber-error flip under fixed boundary conditions

TABLE I
DECISION CHANGES AND PASSES-AND-WRONG BY ERROR TYPE, PROTOCOL AND BED STRUCTURE

Error	Protocol	Discrete bed (97 instances)			Leaky bed (150 instances)		
		<i>n</i>	Flip, %	Passes and wrong, %	<i>n</i>	Flip, %	Passes and wrong, %
T1	A	77	27 (19–38)	13 (7–22)	118	18 (12–26)	16 (11–24)
	B	77	9 (4–18)	1 (0–7)	118	5 (2–11)	5 (2–11)
	C	44	18 (10–32)	20 (11–35)	71	8 (4–17)	11 (6–21)
	D	44	2 (0–12)	23 (13–37)	71	6 (2–14)	4 (1–12)
T2	A	60	40 (29–53)	12 (6–22)	147	43 (35–51)	24 (17–33)
	B	96	18 (11–27)	2 (0–9)	149	5 (2–9)	4 (2–10)
	C	60	20 (12–32)	18 (11–30)	100	8 (4–15)	6 (3–12)
	D	60	12 (6–22)	18 (11–30)	100	6 (3–12)	6 (3–12)
T3	A	97	7 (4–14)	0 (0–4)	150	2 (1–6)	0 (0–2)
	B	97	7 (4–14)	0 (0–4)	150	2 (1–6)	0 (0–2)
	C	97	7 (4–14)	0 (0–4)	150	3 (1–7)	1 (0–4)
	D	97	7 (4–14)	0 (0–4)	150	3 (1–7)	0 (0–2)
T4	A	97	13 (8–22)	6 (3–13)	150	9 (6–15)	7 (4–12)
	B	97	8 (4–15)	0 (0–4)	150	1 (0–4)	0 (0–2)
	C	97	8 (4–15)	9 (5–17)	150	5 (3–10)	24 (18–31)
	D	97	14 (9–23)	36 (27–46)	150	5 (2–9)	8 (5–13)

A flip is a change of classification at FFR 0.80. Passes and wrong means a territory-perfusion residual below 10% and $|\Delta\text{FFR}| > 0.05$, over models with a defined residual (T2 under B in the discrete bed, $n = 60$; T2 under A and B in the leaky bed, $n = 100$). Protocol D matched every territory flow within 1% in all but two models. Wilson 95% intervals in parentheses. The expected flip rate from repeat invasive measurement is 5.0–6.5% across cells.

ended beyond it, whereas 20% (discrete) and 23% (leaky) of topological-error models did (Supplementary Material).

Re-deriving the boundary conditions (Protocol B) produced the fewest flips. Against Protocol A it reversed 14–56 topological flips per cell and created none (McNemar, Holm-adjusted $p < 0.001$), except for discrete-bed vessel breaks (22 reversed, 12 created, $p = 0.24$; mean $\Delta\text{FFR} -0.039$). In the discrete bed, however, Protocol B models failed their own perfusion check most often.

B. Passes-and-Wrong Under Tuned Boundary Conditions

Fitting one bed scaling to the clean territory flows (Protocol C) lowered the median perfusion residual of topological-error models relative to Protocol A from 0.22 to 0.14 in the discrete bed and from 0.11 to 0.02 in the leaky bed (same instances), and reversed more topological flips than it created relative to Protocol A (significant for leaky vessel breaks, $p < 0.001$). Relative to Protocol B, it changed at most two topological-error decisions per cell ($p \geq 0.5$). Protocol D matched the territory flows within 1% in all but two models and corrected most of the topological error: the median $|\Delta\text{FFR}|$ of topological-error models fell to 0.005 in both beds, and their flip rate to 8% (discrete) and 6% (leaky).

Tuning did not remove the error from every model (Fig. 3). After Protocol C, 20 of 104 topological-error models (19%, 13–28%) in the discrete bed and 14 of 171 (8%, 5–13%) in the leaky bed passed the perfusion check while their FFR was materially wrong (more than 0.05 from the clean value); after Protocol D, 21 of 104 (20%, 14–29%) and 9 of 171 (5%, 3–10%) did. On the same instances the proportions were 2% and 6% under Protocol B, whose models passed in 22% and 89%, and 13% and 22% under Protocol A. Tuning raised them relative to B in the discrete bed (McNemar $p < 0.001$ for C and D) but not in the leaky bed ($p \geq 0.12$); relative

to A, Protocol C did not change them in the discrete bed ($p = 0.07$) and lowered them in the leaky bed ($p < 0.001$). Models with correct anatomy, tuned to perfusion measured with physiological noise, passed while materially wrong in 3.1% (2.4–4.0%) and 3.9% (3.2–4.6%) of draws, and their flip rates were 6.2% and 7.2%. Looser thresholds of 13% and 16% raised the proportions (Supplementary Material).

Tuning also turned caliber errors into passes-and-wrong. With re-derived boundary conditions no caliber-error model passed while materially wrong in either bed. After tuning, 5% (C) and 18% (D) did so in the discrete bed and 12% and 4% in the leaky bed (McNemar against B, $p \leq 0.004$ in all four comparisons). Almost all were taper models whose FFR tuning lowered: forcing the clean flow through a uniformly narrowed lumen raises its pressure drop (Table I).

C. Side-Branch Loss Under Fixed and Tuned Resistance

A missed side branch raised the computed FFR. In the 44 (discrete) and 71 (leaky) instances in which tuning was defined, it raised FFR by a median of 0.095 and 0.047 with fixed boundary conditions. Protocol C reduced the shift to 0.077 and 0.015 (Wilcoxon signed-rank, $p < 0.001$ in both beds; Fig. S3 in the Supplementary Material), and Protocol D to 0.000 and 0.004, but 10 of 44 and 3 of 71 instances remained above 0.05 after Protocol D.

D. Three-Dimensional Case Study

The 3D case reproduced the reduced-order result. With the clean tree's resistances at the outlets, removal of the side branch raised the 3D FFR at the measurement point from 0.870 to 0.944 ($\Delta\text{FFR} = +0.074$). With the clean territory flows prescribed at the outlets (the limit of Protocol D), the same error changed FFR by -0.0007 (0.892 in both geometries), so in this instance per-territory tuning removed the error. The reduced-order counterparts gave $+0.081$ and -0.0005 for the

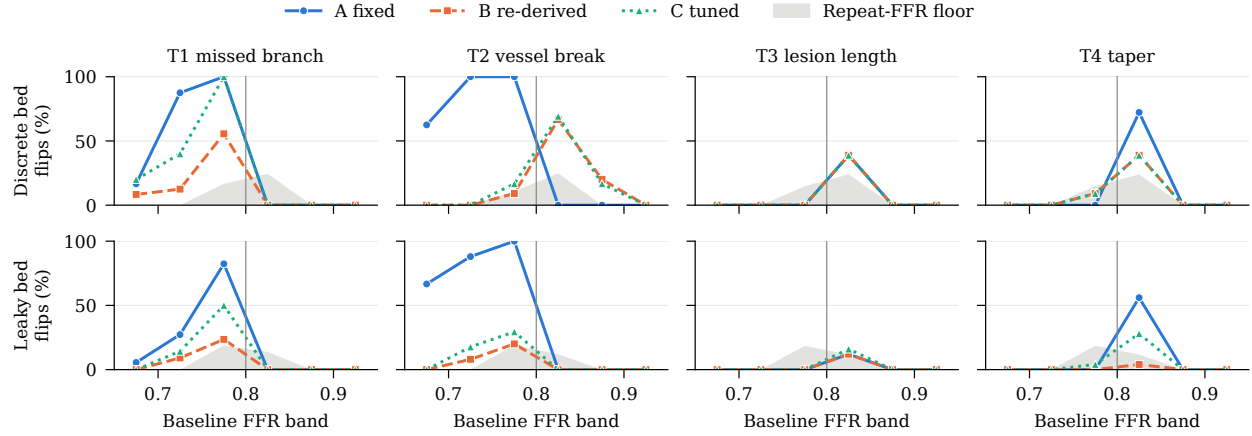


Fig. 2. Flip rate (proportion of models whose classification at FFR 0.80 changed) by baseline FFR band (0.05 wide, plotted at the band center; bands with fewer than three models omitted), error type, protocol (A–C) and bed structure. The shaded area is the flip rate expected from repeat invasive FFR alone (standard deviation 0.018).

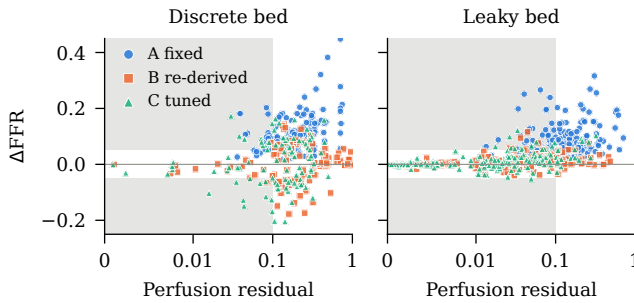


Fig. 3. Change in FFR against territory-perfusion residual for missed-branch and vessel-break models under Protocols A–C (Protocol D residuals lie below 0.01). Each point is one model. The shaded areas hold models that pass the perfusion check (residual < 0.10) while materially wrong ($|\Delta\text{FFR}| > 0.05$). The horizontal axis is linear below 0.01 and logarithmic above.

same two conditions, and their baseline FFR was within 0.011 of the 3D values (Fig. 4). In the cohort model of the same instance (discrete bed), the error gave +0.127 under Protocol A and -0.0007 under Protocol D, whereas the single scaling of Protocol C left +0.107 with a residual of 0.30, which fails the check. Two caveats apply. The meshed lumen was wider than the radius used by the reduced-order model (median difference 0.14 mm; throat area-equivalent radius 0.276 against 0.231 mm), and on the requested radius the reduced-order baseline was 0.761. A finer throat zone (12.5 μm) changed the baseline FFR by at most 0.0021, against the error effect of 0.074 (Supplementary Material).

E. Hyperemic Demand

With k doubled and tripled, the sweep, cohort selection and all four protocols were repeated (Supplementary Material). The median FFR of a 50% DS lesion fell from 0.931 to 0.845 and 0.776, and fewer trees met the discrete-bed criteria (47 and 17 instances; the discrete arm is reported at twice the demand only). With fixed boundary conditions, topological errors changed 33–50% of decisions against 1–5% for caliber

errors; tuning reduced topological flips to 3–20%; and after tuning, 3–20% of topological-error and 1–14% of caliber-error models passed while materially wrong, against 2–5% and 0% with re-derived boundary conditions. These directions held in both beds at twice and in the leaky bed at three times the demand; at twice the demand the excess over re-derived boundary conditions was significant in three of four caliber comparisons and in no topological one.

IV. DISCUSSION

A. What a Perfusion Check Shows After Tuning

Three findings stand out. First, with fixed boundary conditions the decision risk of segmentation lay mainly in the branching structure of the lumen: a topological error changed the decision at 0.80 three to six times as often as a caliber error of inter-observer magnitude, whose flip rates were of the order of repeat invasive measurement. This ranking holds for the magnitudes studied, a whole branch or vessel segment against inter-observer caliber disagreement.

Second, re-deriving or tuning the bed reduced topological flips from 18–43% to 2–20% per error type; per-territory tuning restored the flow that the missing branch had carried through the stenosis, in the cohort and in the 3D case.

Third, after tuning a passing perfusion check did not show that the FFR was correct. Per-territory tuning passes almost every model by design, and in the discrete bed one in five tuned topological-error models passed while materially wrong under either form of tuning, half of those passing under one global scaling. In both beds tuning produced passing models with caliber errors that re-derived models did not, because restoring the clean flow through a narrowed lumen enlarges its pressure drop. The check measured agreement with the fitted targets, not the correctness of the model.

Bed structure set the magnitude: in the leaky bed a deleted branch's conductance reappears as wall outflow at its parent node, damping the error. Claims are therefore limited to directions that held in both; the excess of passes-and-wrong over re-derived boundary conditions for topological errors held only in the discrete bed.

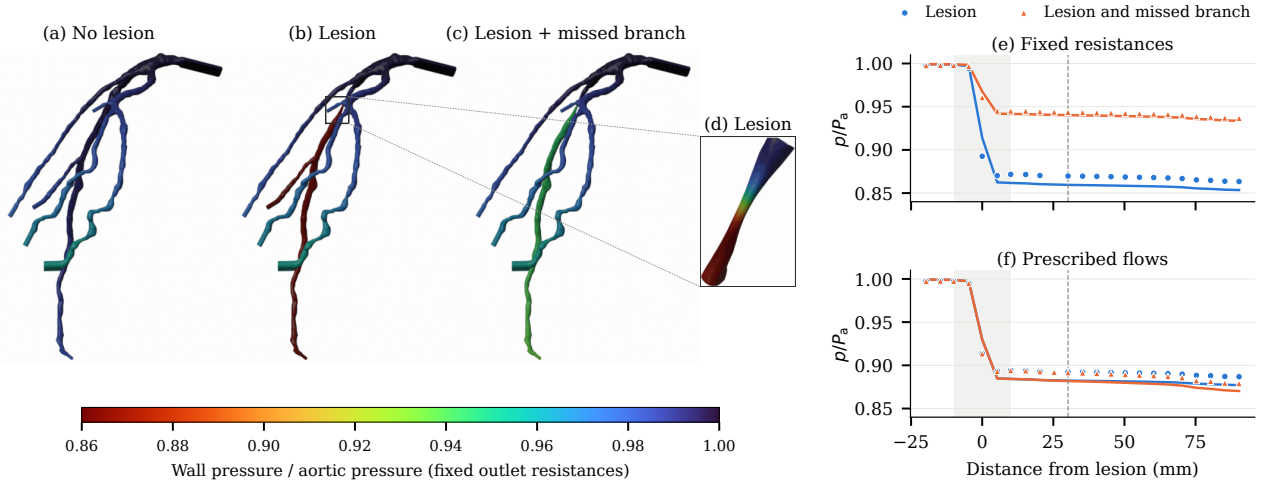


Fig. 4. Three-dimensional case study (proximal LAD, 80% DS). (a)–(c) Wall pressure relative to aortic pressure with the clean tree’s outlet resistances, without the lesion, with the lesion, and with the lesion and a missed side branch. The lesion lowers the downstream pressure (red); the missed branch raises it again (green), a shift that would move a decision near 0.80. (d) Close-up of the lesion boxed in (b). (e), (f) Pressure along the artery with fixed resistances and with prescribed clean territory flows. Markers show 3D cross-section averages and lines the reduced-order models on the meshed radius with the same outlet conditions. The shaded band is the lesion and the dashed line the measurement point.

B. Relation to Published Studies

Published sensitivity analyses perturb lumen caliber at fixed branching, and comparisons of geometry with boundary conditions rank them on a continuous output [14], [15], [16]. Our results add that the ranking depends on the error type and the boundary-condition protocol.

Two earlier findings that appeared to disagree are reproduced here: a side-branch effect of 13–15.5% under fixed outlet resistance [20], and little change in diagnostic accuracy after a cohort-wide outlet resistance was recalibrated [21]. Both appear in our cohort, the first with fixed resistance and the second after re-derivation or tuning, so the difference follows from the boundary-condition protocol.

Vessel breaks occur in the output of current segmentation methods despite high overlap scores [18], and a missing side branch removes few voxels, so the error types that carried the decision risk here are the ones that the usual overlap metrics do not measure, although topology-aware measures such as cIDice exist [32].

C. Implications and Practical Use

Agreement with a fitted quantity does not by itself make a model credible. A coronary digital twin calibrated to perfusion also requires verification, uncertainty quantification and applicability [33]. A missed branch that survived tuning raised FFR, biasing the decision toward deferring treatment, whereas a tuned taper error lowered it, biasing the decision toward treatment.

For a user who computes FFR from an image, the results suggest four steps. First, check that the side branches beyond the lesion are present and that no vessel ends early, because overlap scores do not reveal these errors. Second, if the boundary conditions are tuned to perfusion, check the lumen caliber as well: diameter errors of inter-observer size flipped decisions at rates near repeat invasive measurement (the taper

up to 13%), but after tuning they produced passing models with a wrong FFR. Third, record the perfusion mismatch before tuning; in the discrete bed only 22% of re-derived topological-error models passed, although in the leaky bed most did. Fourth, take most care near 0.80, where decision changes concentrated. For a developer, tuning is a trade-off: per-territory tuning corrected most missed branches but turned caliber errors into passing models with a wrong FFR. These steps follow from the results but are not validated decision rules.

D. Limitations

The inserted stenoses are idealized, and the reduced-order model is steady, without autoregulation or heart-rate effects. The missed-branch and vessel-break magnitudes are design choices, and the caliber magnitudes come from inter-observer statistics that likely overstate agreement. Protocols C and D were undefined for most right coronary instances with a topological error, so their results for these errors describe the left coronary tree. Hyperemic demand followed the segmented inlet radius and was below thermodilution values; at doubled and tripled demand, fewer trees met the healthy-network criterion of the discrete bed. The 3D analysis covers one instance, with a meshed lumen wider than the reduced-order radius; this offset moved the baseline across 0.80 (0.761 against 0.870) and changed the size of the error effect (+0.127 against +0.081), but not its direction or its removal by prescribed flows. No invasive FFR was available, so the noise floor was modeled from published repeat measurements.

V. CONCLUSION

This study proposes a controlled test that introduces one segmentation error at a time into a coronary flow model and judges its effect by the treatment decision at FFR 0.80. Applied to 150 stenoses, it showed that a model that matches perfusion

is not necessarily a correct one. Missed branches and vessel breaks changed the decision three to six times as often as caliber errors of inter-observer size, and tuning the boundary conditions to perfusion reduced these changes without making the models correct: up to one in five tuned models passed a perfusion check stricter than measurement repeatability while their FFR was materially wrong, against 3–4% for correct anatomy. We have shown where safeguards are most effective: a check of the branching near the lesion, which overlap scores miss, and a record of the perfusion mismatch before tuning, which tuning erases. The same test can be applied to any segmentation and boundary-condition pipeline, including those that calibrate coronary digital twins to perfusion.

DATA AND CODE AVAILABILITY

The data that support the findings of this study were derived from resources available in the public domain: ImageCAS [22] and ImageCAS-X [23] (CC BY 4.0). The code is available from the corresponding author upon reasonable request.

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